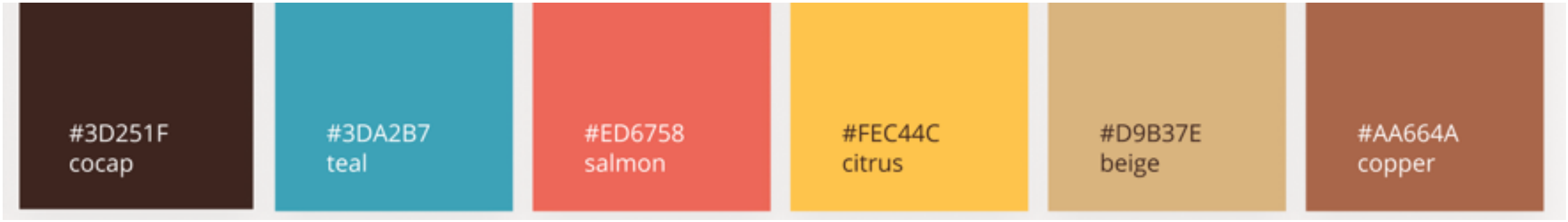


Dramatically improves scalability

smooth & differentiable

Sinkhorn Divergence (SD)



#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

Implications of EOT

- Computational:
 - **Dramatically improves scalability**: from cubic LP solvers to near-quadratic Sinkhorn iterations.
 - GPU-friendly: simple matrix–vector updates, parallelizable.
 - Handles large datasets (millions of points) that classical OT can't.
- Optimization-friendly:
 - Entropic regularization makes OT loss **smooth & differentiable**.
 - Gradients propagate through Sinkhorn iterations → usable as a loss in deep nets.
 - Enables end-to-end learning with OT layers.
- Statistical:
 - Defines the **Sinkhorn Divergence (SD)**: unbiased correction of entropic OT cost.
 - Balances comp. efficiency with better statistical convergence than classical Wasserstein in high dim
 - We will talk about statistical properties more later on

Applications in ML